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1.) Check quality with fastqc, but using falco (Galaxy Version 1.3.0+galaxy0) b/c faster
    - Default parameters
    - Run with transcriptome reads
        - Forward data
            - Fails: per tile sequence quality; Sequence duplication levels; Adapter Content
            - Warnings: Per base sequence content; Per sequence GC content; Overrepresented sequences
        - Reverse Data
            - Fails: Per tile sequence quality; adapter content
            - Warnings: Per base sequence content; Per sequence GC content; Sequence duplication levels; Overrepresented sequences

2.) Trim with fastp (Galaxy Version 1.3.3+galaxy0)
    - Set to paired collection
    - Gets rid of stuff that are fails for the quality control
        - Observe the results on HTML, those under filtering reports

3.) HISAT2 (Galaxy Version 2.2.2+galaxy0) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar
    - Change in parameters:
- Source for reference genome: genome from the history
- Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ((Galaxy Version 2.2.3+galaxy0) - assembles short reads into actual transcripts, predicts full length transcripts
- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts
- Change parameter to reference genome to history
    - Burkholderia.gff

Step 4.) StringTieMerge (Galaxy Version 2.2.3+galaxy0) - instead of 6 files, puts them all together
- Use dataset collection and use the StringTie file
- Use reference as burkholderia.gff

- step 5.) featureCounts (Galaxy Version 2.1.1+galaxy0)
- Set alignment file to dataset collection and use HISAT file
- Set gene annotation file as GFF/GTF from history and use latest stringtiemerge
- Rewrite GFF gene identifier as transcript_id

- Does the input have read pairs? Set to Yes, paired-end and count them as 1 single fragment
    - Deseq2 (Galaxy Version 2.11.40.8+galaxy2) - function? Takes all 6 conditions and sees if it is more represented in one condition or the other, gives statistical significance
- Factor: set to Growth_Media
- 2 factor levels. Factor level 1 set as Broth; Factor level 2 set as Plate
    - Rename the counts as wildtype or mutants
    - Tells if gene was up or down regulated
    - Use beta priors? Set to yes
    - Output options: generate plots and output normalised counts
    - Click “include hidden” within history and locate the featurecount and find the Counts, rename as plate/BrothReplicate#
    - Put these in using the multiple dataset options, search with the ellipses
- From the result file you get a p-value, can determine null hypothesis
    - For the gene, there is no change in gene expression between plate and broth: null hypothesis
    - If there are changes between the two growth media, you can infer significant difference, but need this p-value to make a statistical backing for your reasoning
    - Less than 0.05, it is statistically significant